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The AI Project Discovery & Spec-Driven Design Master Prompt

Version: 1.1 **Date:** June 2026

1. What This Prompt Does

This is a **7-phase, role-stacked mega-prompt** that transforms a raw project idea into a complete, implementation-ready documentation package. It instructs an AI to act simultaneously as:

Senior Business Analyst · Enterprise Solution Architect · Technical Product Manager
· Software Architect · DevOps Architect · Security Architect · AI Engineering Lead

Rather than asking for a single document, the prompt **chains 7 sequential phases**, each gated by the previous, and produces **9 specification artifacts** that collectively drive both human engineers and AI coding assistants.

2. Prompt Structure — Phase by Phase

Phase 1 — Requirements Discovery

Instruction: Ask interactive questions. Do NOT generate documents yet. Group questions by category. Only ask highest-priority unanswered questions.

Categories covered:

- Business Information (goals, stakeholders, budget, timeline)
- Functional Requirements (features, roles, workflows, integrations)
- Non-Functional Requirements (scalability, security, compliance, performance)
- AI Requirements (LLM, RAG, agents, fine-tuning, evaluation)
- Deployment Requirements (cloud, on-premise, DR, backup)

Output: A complete set of answered requirements — no artifact yet.

Phase 2 — BRD Generation

Instruction: Generate `/artifacts/BRD.md` using professional enterprise-grade formatting.

Sections mandated: Executive Summary · Business Objectives · Scope (In/Out) · Stakeholders · User Personas · Functional Requirements · Non-Functional Requirements · Assumptions · Risks · Dependencies · Acceptance Criteria · Success Metrics · Roadmap

Output: `BRD.md`

Phase 3 — Architecture Discovery

Instruction: Ask interactive architecture questions. Cover style, frontend, backend, database, AI components, infrastructure, security, observability, CI/CD.

Decisions forced:

- Architecture style (monolith → microservices → serverless)
- Framework selection (frontend + backend)
- API style (REST / GraphQL / gRPC)
- Database type (SQL / NoSQL / Vector / Time Series)
- LLM + embedding model + vector DB choices
- Auth / secrets / SSO / MFA
- Logging / monitoring / tracing / alerting
- Git strategy + CI/CD tooling

Output: A complete set of answered architecture decisions — no artifact yet.

Phase 4 — System Design Documents (5 artifacts)

Instruction: Generate 5 markdown files with Mermaid diagrams wherever possible.

Artifact	Key Contents
SystemArchitecture.md	Context · Container · Component · Deployment · Data Flow · Integration · Security · HA · Scalability diagrams
DatabaseDesign.md	ERD · Table definitions · Index strategy · Partitioning · Retention · Mermaid ER diagrams
API_Specification.md	All endpoints · Request/Response · Validation · Auth · Error handling · OpenAPI-compatible
SecurityDesign.md	Threat model · STRIDE · Security controls · IAM · Encryption · Audit logging · Compliance
DevOpsDesign.md	CI/CD architecture · Environments · Deployment · Rollback · Monitoring · Logging · DR
TestStrategy.md	Unit · Integration · E2E · Performance · Security · UAT strategies

Output: 6 artifacts (SystemArchitecture + 5 supporting docs)

Phase 5 — Project Execution Plan

Instruction: Generate `/artifacts/ExecutionPlan.md` with full implementation roadmap.

Required elements: Milestones · Epics · Features · Tasks · Dependencies · Risks · Resource Requirements · Estimated Effort · Delivery Sequence · Critical Path

Output: `ExecutionPlan.md`

Phase 6 — AI Coding Assistant Implementation Prompt

Instruction: Generate `/artifacts/AI_Implementation_Prompt.md` optimised for Cursor, Claude Code, Cline, Roo Code, Windsurf, GitHub Copilot Agent, OpenAI Codex.

Required sections: Project Context · Referenced Documents · Development Rules · Coding Standards · Architecture Constraints · Implementation Phases · Validation Gates · Testing Requirements · Deliverables · Definition of Done

Output: `AI_Implementation_Prompt.md`

Phase 7 — Artifact Management

Instruction: Maintain consistency across all artifacts. When information is missing, ask questions and update impacted documents. Never assume critical requirements.

Ongoing rules:

- All outputs as markdown files
 - Maintain `/artifacts/` directory structure
 - Cross-update when any artifact changes
 - Never assume — always validate
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3. Full Artifact Output Map

/artifacts/

— BRD.md	← Phase 2
— SystemArchitecture.md	← Phase 4
— DatabaseDesign.md	← Phase 4
— API_Specification.md	← Phase 4
— SecurityDesign.md	← Phase 4
— DevOpsDesign.md	← Phase 4
— TestStrategy.md	← Phase 4
— ExecutionPlan.md	← Phase 5
— AI_Implementation_Prompt.md	← Phase 6

Total: 9 files
